

BEE 4750 Homework 4: Linear Programming and Capacity Expansion

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Due Date

Thursday, 11/06/25, 9:00pm

Overview

Instructions

- Problem 1 asks you to analytically (or graphically) solve a linear program.
- Problem 2 asks you to formulate and solve a resource allocation problem using linear programming.
- Problem 3 asks you to formulate, solve, and analyze a standard generating capacity expansion problem.

Load Environment

The following code loads the environment and makes sure all needed packages are installed. This should be at the start of most Julia scripts.

```
import Pkg
Pkg.activate(@__DIR__)
Pkg.instantiate()
```

Activating project at `~/hw4-ero23\_hw4`

```
using JuMP
using HiGHS
using DataFrames
using Plots
using Measures
using CSV
using MarkdownTables
```

Problems (Total: 30 Points)

Problem 1 (Total: 3 Points)

Solve the following linear program **analytically**. Show your work and justify any reasoning.

$$\begin{aligned} \max_{x,y} \quad & 3x + 2y \\ \text{subject to:} \quad & x + 4y \leq 16 \\ & x - 2y \leq -1 \\ & 0 \leq x \leq 4 \\ & 1 \leq y \leq 4. \end{aligned}$$

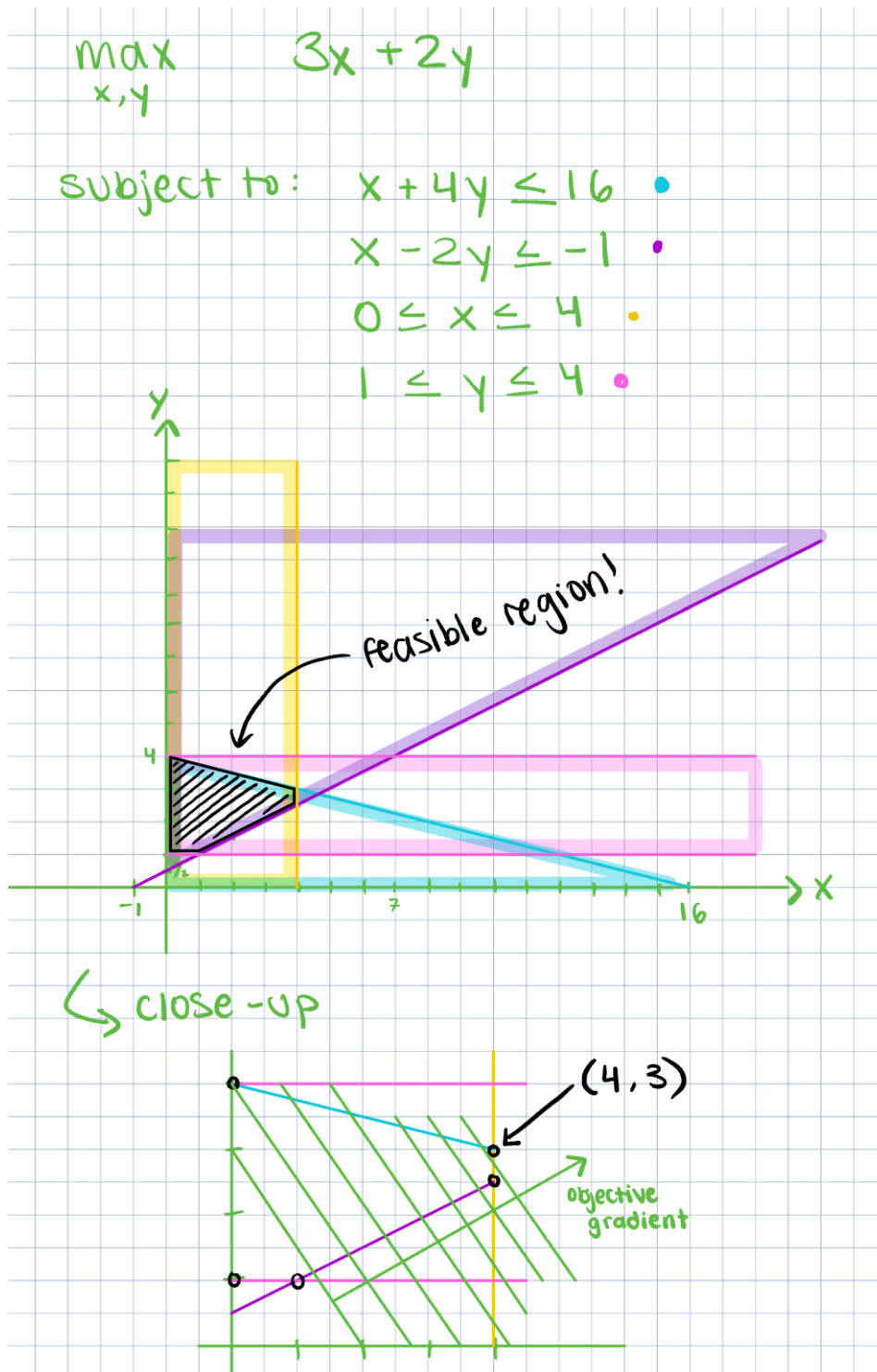


Figure 1: Hand-drawn Graph of LP problem

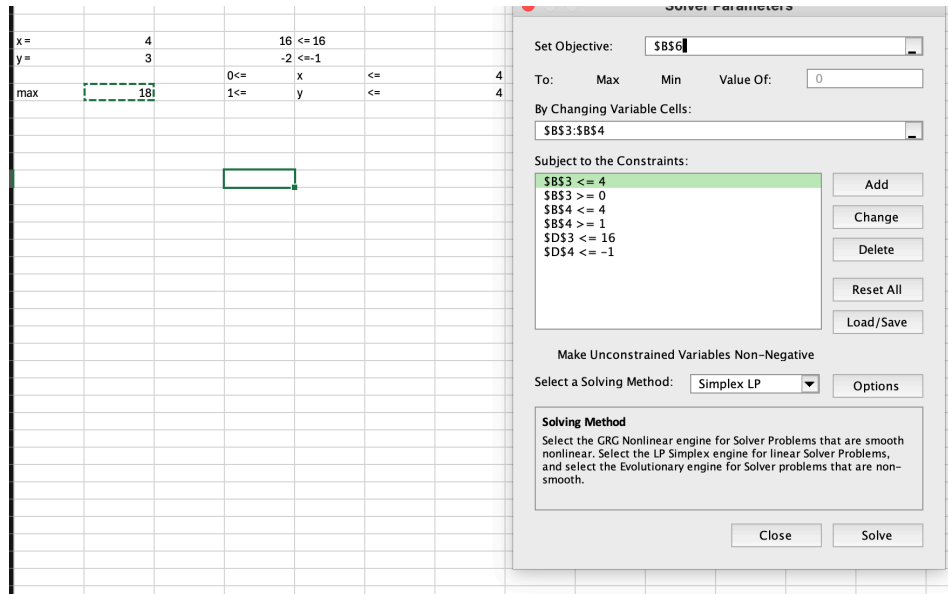


Figure 2: Checked the analytic solution using excel solver

Problem 2 (12 points)

A farmer has access to a pesticide which can be used on corn, soybeans, and wheat fields, and costs \$70/kg-yr to apply. The crop yields the farmer can obtain following crop yields by applying varying rates of pesticides to the field are shown in Table 1.

Application Rate (kg/ha)	Soybean (kg/ha)	Wheat (kg/ha)	Corn (kg/ha)
0	2900	3500	5900
1	3800	4100	6700
2	4400	4200	7900

Table 1: Crop yields from applying varying pesticide rates for Problem 1.

The costs of production, *excluding pesticides*, for each crop, and selling prices, are shown in Table 2.

Crop	Production Cost (\$/ha-yr)	Selling Price (\$/kg)
Soybeans	350	0.36
Wheat	280	0.27
Corn	390	0.22

Table 2: Costs of crop production, excluding pesticides, and selling prices for each crop.

Recently, environmental authorities have declared that farms cannot have an *average* application rate on soybeans, wheat, and corn which exceeds 0.8, 0.7, and 0.6 kg/ha, respectively. The farmer has asked you for advice on how they should plant crops and apply pesticides to maximize profits over 130 total ha while remaining in regulatory compliance if demand for each crop (which is the maximum the market would buy) this year is 250,000 kg?

Problem 2.1

Formulate a linear program for this resource allocation problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations). **Tip: Make sure that all of your constraints are linear.**

```
farm_model = Model(HiGHS.Optimizer) # initialize model object
@variable(farm_model, S[0:2] >= 0) # non-negativity constraints
@variable(farm_model, W[0:2] >= 0)
@variable(farm_model, C[0:2] >= 0)

# Maximize profit
@objective(farm_model, Max, 694S[0]+948S[1]+1094S[2]+665W[0]+757W[1]+714W[2]+908C[0]+1014C[1]+1014C[2])

# Total land constraint
@constraint(farm_model, land, S[0]+S[1]+S[2]+W[0]+W[1]+W[2]+C[0]+C[1]+C[2] <= 130)

# Demand (must not exceed market limit)
@constraint(farm_model, soybean_max, 2900S[0]+3800S[1]+4400S[2] <= 250_000)
@constraint(farm_model, wheat_max, 3500W[0]+4110W[1]+4200W[2] <= 250_000)
@constraint(farm_model, corn_max, 5900C[0]+6700C[1]+7900C[2] <= 250_000)

# Pesticide average constraints (linearized)
@constraint(farm_model, -0.8 * S[0] + 0.2 * S[1] + 1.2 * S[2] <= 0)
@constraint(farm_model, -0.7 * W[0] + 0.3 * W[1] + 1.3 * W[2] <= 0)
@constraint(farm_model, -0.6 * C[0] + 0.4 * C[1] + 1.4 * C[2] <= 0)

print(farm_model) # this outputs a nicely formatted summary of the model so you can check your work
```

Max 694 S[0] + 948 S[1] + 1094 S[2] + 665 W[0] + 757 W[1] + 714 W[2] + 908 C[0] + 1014 C[1] + 1014 C[2]
 Subject to
 land : S[0] + S[1] + S[2] + W[0] + W[1] + W[2] + C[0] + C[1] + C[2] 130
 soybean_max : 2900 S[0] + 3800 S[1] + 4400 S[2] 250000

```

wheat_max : 3500 W[0] + 4110 W[1] + 4200 W[2]    250000
corn_max  : 5900 C[0] + 6700 C[1] + 7900 C[2]    250000
-0.8 S[0] + 0.2 S[1] + 1.2 S[2]    0
-0.7 W[0] + 0.3 W[1] + 1.3 W[2]    0
-0.6 C[0] + 0.4 C[1] + 1.4 C[2]    0
S[0]    0
S[1]    0
S[2]    0
W[0]    0
W[1]    0
W[2]    0
C[0]    0
C[1]    0
C[2]    0

```

```
optimize!(farm_model)
```

Running HiGHS 1.11.0 (git hash: 364c83a51e): Copyright (c) 2025 HiGHS under MIT licence terms

LP has 7 rows; 9 cols; 27 nonzeros

Coefficient ranges:

Matrix [2e-01, 8e+03]

Cost [7e+02, 1e+03]

Bound [0e+00, 0e+00]

RHS [1e+02, 2e+05]

Presolving model

7 rows, 9 cols, 27 nonzeros 0s

7 rows, 9 cols, 27 nonzeros 0s

Presolve : Reductions: rows 7(-0); columns 9(-0); elements 27(-0) - Not reduced

Problem not reduced by presolve: solving the LP

Using EKK dual simplex solver - serial

Iteration	Objective	Infeasibilities	num(sum)
0	-1.8170297740e+02	Ph1: 7(28.3649); Du: 9(181.703)	0s
7	1.1674116702e+05	Pr: 0(0)	0s

Model status : Optimal

Simplex iterations: 7

Objective value : 1.1674116702e+05

P-D objective error : 6.2325284535e-17

HiGHS run time : 0.00

Problem 2.2

How many ha should the farmer dedicate to each crop and with what pesticide application rate(s)? How much profit will the farmer expect to make?

```
println("=== Optimal Land Allocation (hectares) ===")

println("\nSoybeans:")
println("  S[0] (0 kg/ha pesticide) = ", value(S[0]))
println("  S[1] (1 kg/ha pesticide) = ", value(S[1]))
println("  S[2] (2 kg/ha pesticide) = ", value(S[2]))

println("\nWheat:")
println("  W[0] (0 kg/ha pesticide) = ", value(W[0]))
println("  W[1] (1 kg/ha pesticide) = ", value(W[1]))
println("  W[2] (2 kg/ha pesticide) = ", value(W[2]))

println("\nCorn:")
println("  C[0] (0 kg/ha pesticide) = ", value(C[0]))
println("  C[1] (1 kg/ha pesticide) = ", value(C[1]))
println("  C[2] (2 kg/ha pesticide) = ", value(C[2]))
```

=== Optimal Land Allocation (hectares) ===

Soybeans:

S[0] (0 kg/ha pesticide) = 13.812154696132604
S[1] (1 kg/ha pesticide) = 55.24861878453038
S[2] (2 kg/ha pesticide) = 0.0

Wheat:

W[0] (0 kg/ha pesticide) = 6.7433064173395625
W[1] (1 kg/ha pesticide) = 15.734381640458977
W[2] (2 kg/ha pesticide) = 0.0

Corn:

C[0] (0 kg/ha pesticide) = 26.92307692307692
C[1] (1 kg/ha pesticide) = 0.0
C[2] (2 kg/ha pesticide) = 11.538461538461547

```
println("\nTotal Maximum Profit = \", objective_value(farm_model))
```

Total Maximum Profit = \$116741.16702082448

Problem 2.3

The farmer has an opportunity to buy an extra 10 ha of land. How much extra profit would this land be worth to the farmer? Discuss why this value makes sense and under what conditions you would recommend the farmer should make the purchase.

```
println("Shadow price of land constraint = ", shadow_price(land))  
  
println("Extra profit for 10 extra ha= $", shadow_price(land)*10)
```

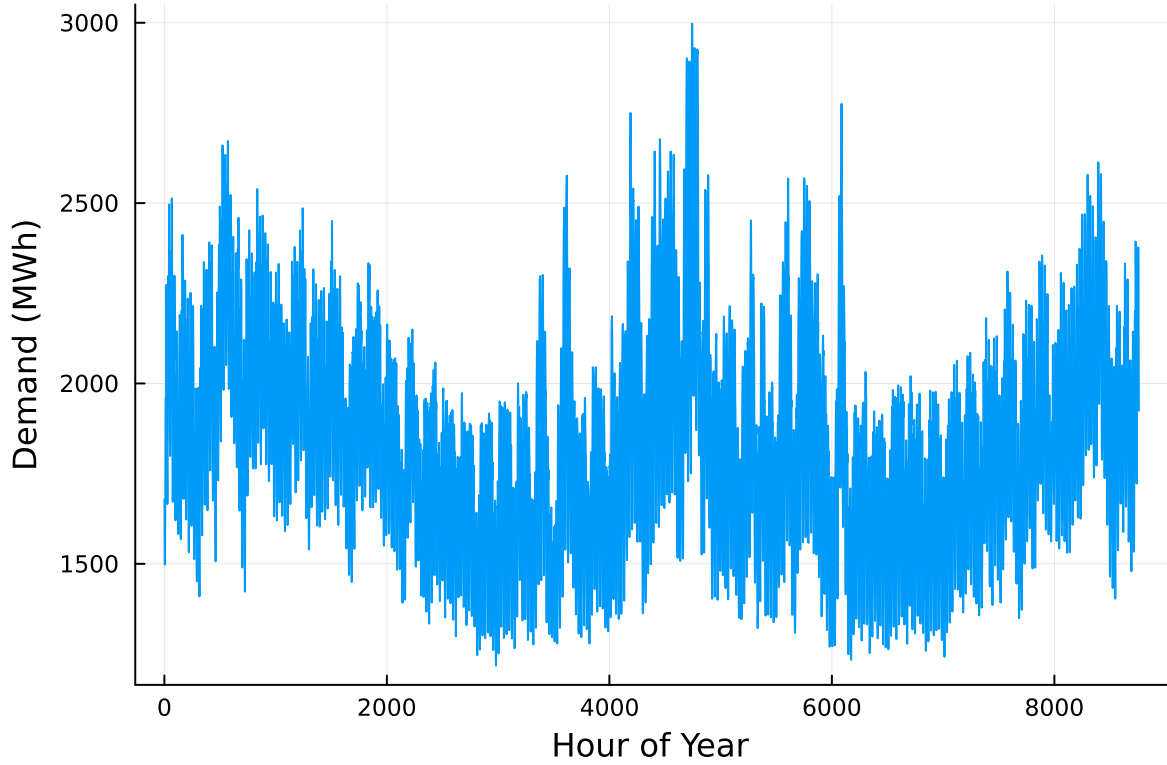
```
Shadow price of land constraint = 729.4  
Extra profit for 10 extra ha= $7294.0
```

This value is based on the shadow price for 1 ha, and then multiplied by 10. Given the profit from the 130 ha was \$116,741.17, each 10 ha equals approximately \$8,980 which aligns with the price we got. I would recommend the farmer purchases the land if the price for the land is less than the \$7294.0.

Problem 3 (15 points)

For this problem, we will use hourly load (demand) data from 2013 in New York's Zone C (which includes Ithaca). The load data is loaded and plotted below in Figure 1.

```
# load the data, pull Zone C, and reformat the DataFrame  
NY_demand = DataFrame(CSV.File("data/2013_hourly_load_NY.csv"))  
rename!(NY_demand, "Time Stamp" => :Date)  
demand = NY_demand[:, [:Date, :C]]  
rename!(demand, :C => :Demand)  
demand[:, :Hour] = 1:nrow(demand)  
  
# plot demand  
plot(demand.Hour, demand.Demand, xlabel="Hour of Year", ylabel="Demand (MWh)", label=:false)
```

Next, we load the generator data, shown in Table 3. This data includes fixed costs (\$/MW installed), variable costs (\$/MWh generated), and CO2 emissions intensity (tCO2/MWh generated).

```
gens = DataFrame(CSV.File("data/generators.csv"))
```

	Plant	FixedCost	VarCost	Emissions
	String15	Int64	Int64	Float64
1	Geothermal	450000	0	0.0
2	Coal	220000	24	1.0
3	NG CCGT	82000	30	0.43
4	NG CT	65000	40	0.55
5	Wind	91000	0	0.0
6	Solar	70000	0	0.0

Finally, we load the hourly solar and wind capacity factors, which are plotted in Figure 2. These tell us the fraction of installed capacity which is expected to be available in a given hour for generation (typically based on the average meteorology).

```

# load capacity factors into a DataFrame
cap_factor = DataFrame(CSV.File("data/wind_solar_capacity_factors.csv"))

# plot January capacity factors
p1 = plot(cap_factor.Wind[1:(24*31)], label="Wind")
plot!(cap_factor.Solar[1:(24*31)], label="Solar")
axis!("Hour of the Month")
axis!("Capacity Factor")

p2 = plot(cap_factor.Wind[4344:4344+(24*31)], label="Wind")
plot!(cap_factor.Solar[4344:4344+(24*31)], label="Solar")
axis!("Hour of the Month")
axis!("Capacity Factor")

display(p1)
display(p2)

```

You have been asked to develop a generating capacity expansion plan for the utility in Riley County, NY, which currently has no existing electrical generation infrastructure. The utility can build any of the following plant types: geothermal, coal, natural gas combined cycle gas turbine (CCGT), natural gas combustion turbine (CT), solar, and wind. The NY state legislature is planning to enact an annual CO₂ limit, which for the utility would limit the emissions in its footprint to 1.5 MtCO₂/yr.

While coal, CCGT, and CT plants can generate at their full installed capacity, geothermal plants operate at maximum 85% capacity, and solar and wind available capacities vary by the hour depend on the expected meteorology. The utility will also penalize any non-served demand at a rate of \$10,000/MWh.

Problem 3.1

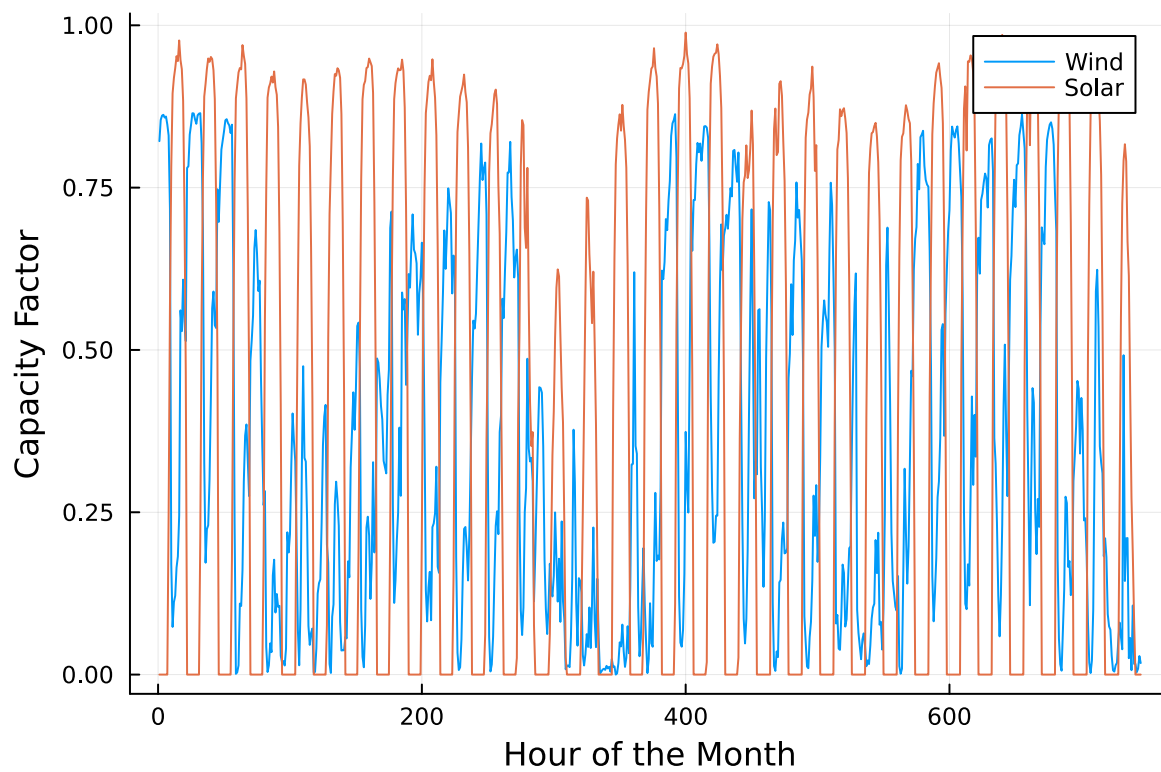
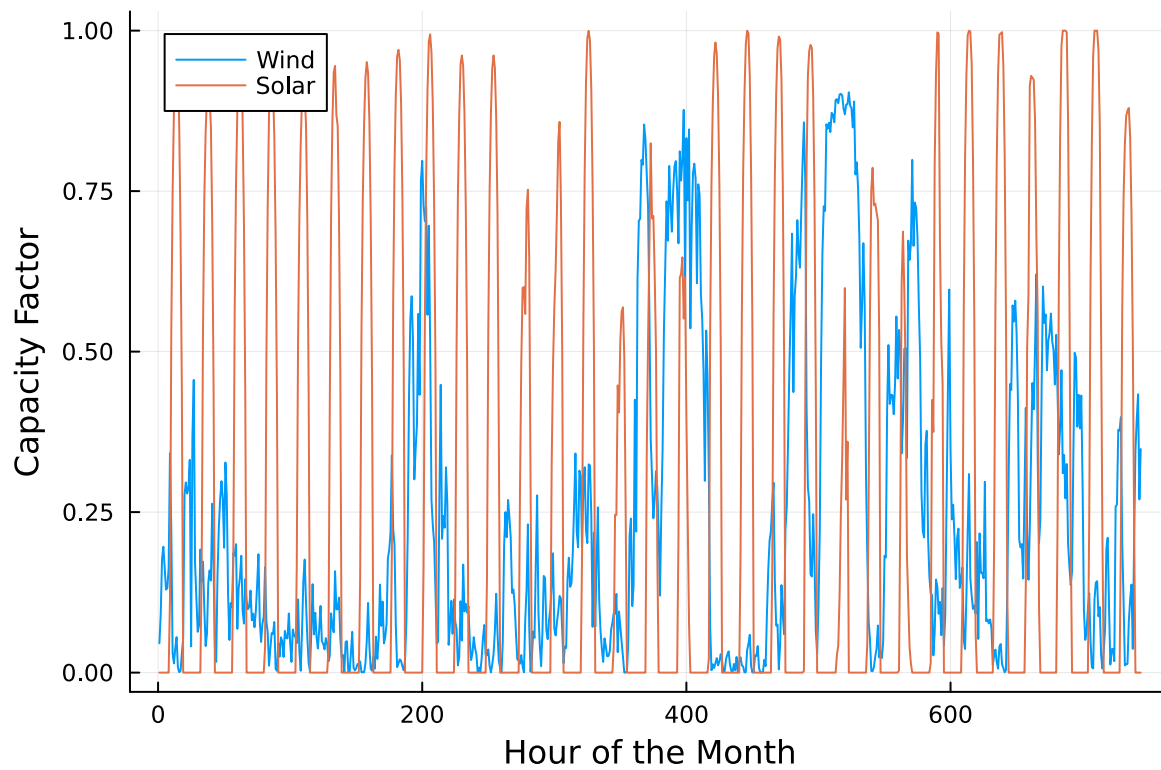
Formulate a linear program for this capacity expansion problem, including clear definitions of decision variable(s) (including units), objective function(s), and constraint(s) (make sure to explain functions and constraints with any needed derivations and explanations).

```

gens = DataFrame(CSV.File("data/generators.csv"))
wind_cap = cap_factor.Wind      # hourly wind capacity factors
solar_cap = cap_factor.Solar     # hourly solar cap. factors

fixed_cost = gens.FixedCost      # $/MW installed
var_cost = gens.VarCost          # $/MWh
co2_rate = gens.Emissions        # tons of CO2 per MWh

```



```

NSE_Cost = 10_000          #$/MWh
G = 1:nrow(gens)          #generators
T = 1:nrow(demand)        #hours in the year
CO2_lim = 1_500_000       #tCO2/year

energy_model = Model(HiGHS.Optimizer)
#define variables
@variables (energy_model, begin
    x[g in G] >= 0          # Installed capacity of generator g (MW)
    y[g in G, t in T] >= 0  # Generation in hour t by g (MWh)
    NSE[t in T] >= 0        # Unmet demand (MWh)
end)

#Minimize costs
@objective (energy_model, Min,
    sum(fixed_cost[g] * x[g] for g in G) +
    sum(var_cost[g] * y[g, t] for g in G, t in T) +
    NSE_Cost * sum(NSE[t] for t in T)
)

#Demand met, or becomes NSE
@constraint (energy_model, load_balance[t in T],
    sum(y[g, t] for g in G) + NSE[t] >= demand.Demand[t]
)

#Capacity limits (Wind, Solar, Geo)
@constraint (energy_model, wind_limit[g in G, t in T; gens[g, :Plant] == "Wind"],
    y[g, t] <= x[g] * wind_cap[t]
)
@constraint (energy_model, solar_limit[g in G, t in T; gens[g, :Plant] == "Solar"],
    y[g, t] <= x[g] * solar_cap[t]
)
@constraint (energy_model, geo_limit[g in G, t in T; gens[g, :Plant] == "Geothermal"],
    y[g, t] <= 0.85 * x[g]
)

# Capacity constraints for dispatchable plants (Coal, CCGT, CT)
@constraint (energy_model, dispatchable_cap[g in G, t in T;
    gens.Plant[g] ["Wind", "Solar", "Geothermal"]],
    y[g, t] <= x[g]
)

```

```
#CO2 Emissions Cap constraint
@constraint(energy_model, co2_cap,
    sum(co2_rate[g] * y[g, t] for g in G, t in T) <= CO2_lim
)

optimize!(energy_model);
```

Running HiGHS 1.11.0 (git hash: 364c83a51e): Copyright (c) 2025 HiGHS under MIT licence terms

LP has 61321 rows; 61326 cols; 188256 nonzeros

Coefficient ranges:

Matrix [5e-05, 1e+00]

Cost [2e+01, 4e+05]

Bound [0e+00, 0e+00]

RHS [1e+03, 2e+06]

Presolving model

56857 rows, 56862 cols, 179328 nonzeros 0s

56854 rows, 56859 cols, 179322 nonzeros 0s

Presolve : Reductions: rows 56854(-4467); columns 56859(-4467); elements 179322(-8934)

Solving the presolved LP

Using EKK dual simplex solver - serial

Iteration	Objective	Infeasibilities	num(sum)
0	0.0000000000e+00	Pr: 8760(3.56131e+06)	0s
26530	7.1866966237e+08	Pr: 7916(894919); Du: 0(5.22508e-06)	5s
30978	7.8436578999e+08	Pr: 1588(45648.3); Du: 0(4.63825e-06)	10s
31987	7.8535295894e+08	Pr: 0(0); Du: 0(8.32377e-11)	11s

Solving the original LP from the solution after postsolve

Model status : Optimal

Simplex iterations: 31987

Objective value : 7.8535295894e+08

P-D objective error : 1.3281687820e-14

HiGHS run time : 11.54

Problem 3.2

How much should the utility build of each type of generating plant? What will the total cost be? How much energy will be non-served?

285.6 MW of Geothermal, 1509.2 MW of CCGT, 703.1 MW of CT, 2548.9 MW Wind, 2103.7 MW Solar.

The total Cost = \$785,352,958.94. Non-served energy = 121 MW

Problem 3.3

What fraction of annual generation does each plant type produce? How does this compare to the breakdown of built capacity that you found in Problem 3.2? Do these results make sense given the demand and generator data?

The percent of annual production for each plant is shown in the final column in the chart below. This aligns somewhat with the breakdown of the built capacity, in that wind and solar have the most built capacity and provide the highest annual percentage. The NG CT plant does not account for even 1% of annual production, while geothermal accounts for almost 7% and has far less installed capacity. This makes sense because of the emissions from the NG CT plant, it is only used when nothing else will provide enough power, whereas the geothermal can run constantly with no emissions.

```
### COPIED FROM CLASS SLIDES TO SUMMARIZE RESULTS ###
```

```
generation = zeros(size(G,1))
for i in 1:size(G,1)
    generation[i] = sum(value.(y)[G[i],:].data)
end
MWh_share = generation./sum(demand.Demand).*100
cap_share = value.(x).data ./ maximum(demand.Demand) .* 100
results = DataFrame(
    Resource = gens.Plant[G],
    Capacity = value.(x).data,
    Percent_Cap = cap_share,
    Generated = generation/1000,
    Percent_Gen = MWh_share
)
rename!(results,
    [:Capacity => : "Installed (MW)",
     :Percent_Cap => : "Percent (%)",
     :Generated => "Generation (GWh)",
     :Percent_Gen => "Percent (%)"]
)
results[:, 2:end] = round.(results[:, 2:end], digits=1)

NSE_MW = maximum(value.(NSE).data)
NSE_MWh = sum(value.(NSE).data)
push!(results, ["NSE" NSE_MW NSE_MW/maximum(demand.Demand)*100 NSE_MWh/1000 NSE_MWh/sum(demand.Demand)*100])

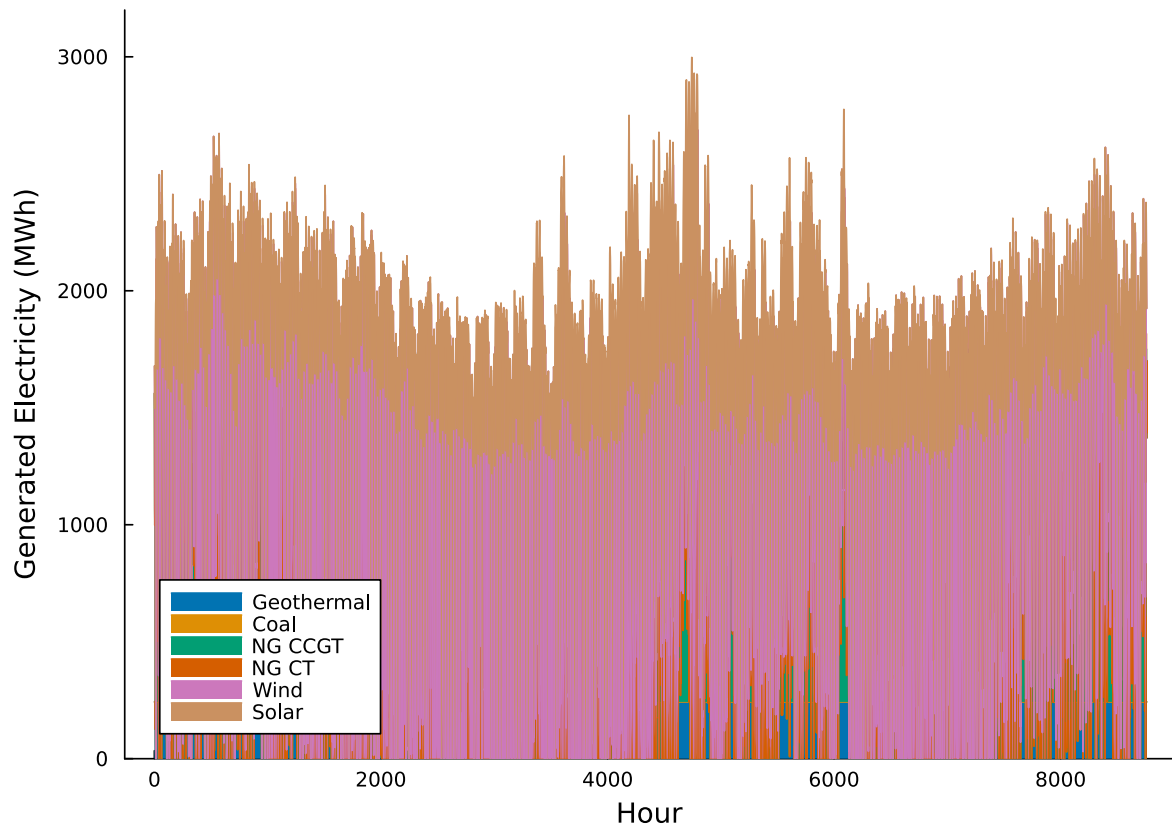
println("\nTotal Cost = \$", objective_value(energy_model))

results[:, 2:end] = round.(results[:, 2:end], digits=1)
markdown_table(results)
```

Total Cost = \$7.853529589434901e8

Resource	Installed (MW)	Percent (%)	Generation (GWh)	Percent (%)
Geothermal	285.6	9.5	1106.3	6.8
Coal	0.0	0.0	0.0	0.0
NG CCGT	1509.2	50.3	3322.8	20.3
NG CT	703.1	23.5	129.5	0.8
Wind	2548.9	85.0	6598.7	40.3
Solar	2103.7	70.2	5210.7	31.8
NSE	121.0	4.0	0.3	0.0

```
### ALSO COPIED FROM CLASS SLIDES ###
p2 = areaplot(value.(y).data',
    label=permutedims(gens.Plant),
    xlabel = "Hour",
    ylabel ="Generated Electricity (MWh)",
    color_palette=:seaborn_colorblind,
    grid=:false,
    legend = :bottomleft)
ylims!(p2, (0, 3200))
plot!(p2, size=(700, 500))
```



Problem 3.5

What is the shadow price of CO₂? What is the value to the utility be of allowing it to emit an additional 1000 tCO₂/yr?

```
println("Shadow price of CO2 = \$", -shadow_price(co2_cap))
println("Value of relaxing CO2 by 1000 tCO2 = \$", -shadow_price(co2_cap)*1000)
```

Shadow price of CO₂ = \$182.77193609185855

Value of relaxing CO₂ by 1000 tCO₂ = \$182771.93609185854

References

I consulted the class slides, and code used there.